Project Report: Zomato Restaurant Data Analysis

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1. Executive Summary

This report provides a data-driven analysis of the Zomato restaurant dataset. The goal was to identify key factors influencing restaurant success, including location density, cuisine popularity, and the relationship between pricing and ratings.

2. Data Cleaning & Methodology

To ensure high-quality insights, the following steps were performed:

- **Standardization:** All column names were normalized to lowercase to prevent mapping errors.
- **Handling Missing Values:** Missing entries in the 'Cuisines' field were labeled as 'Other' to preserve the integrity of the remaining data.
- **Deduplication:** Redundant restaurant entries were removed to prevent bias in location hotspots.
- **Encoding:** Handled specific character encoding (Latin-1) to ensure all international restaurant names were captured correctly.

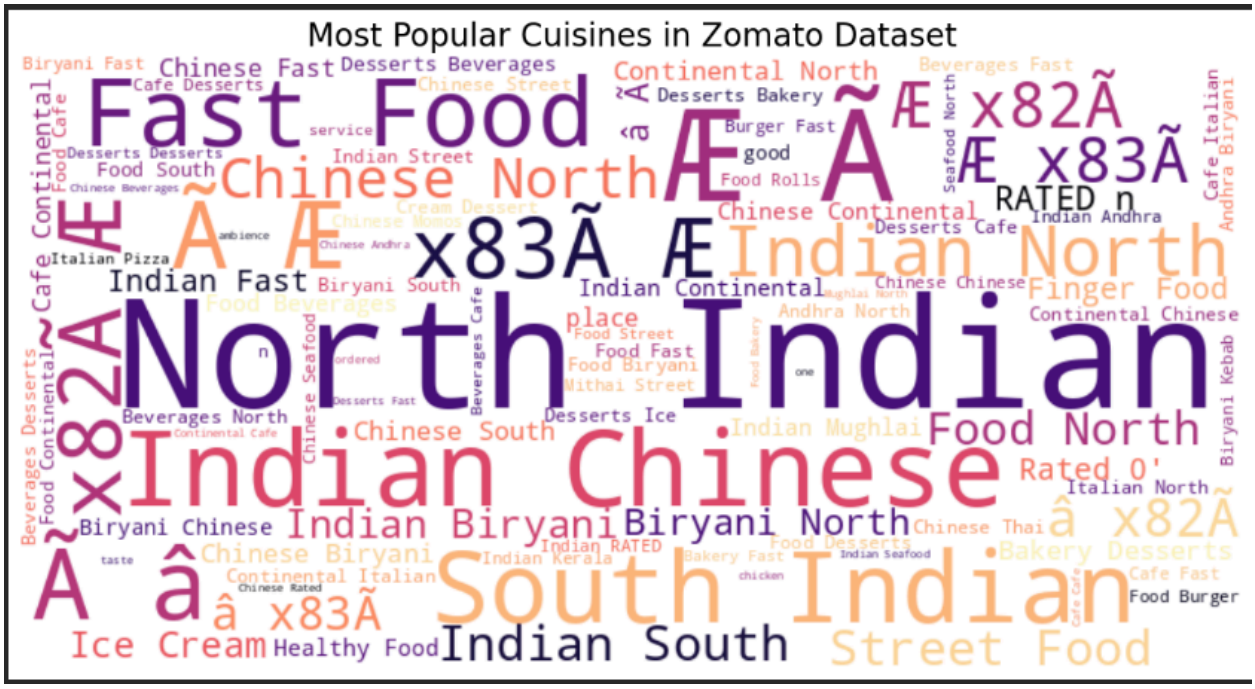
3. Key Finding: Location Hotspots

Our analysis identified the top cities with the highest restaurant density.

- **Insight:** A high concentration of restaurants in specific cities indicates a mature market but also high competition.
- **Visual Evidence:** (Note: Insert your Bar Chart screenshot from Google Colab here).

4. Analysis: Cuisines and Ratings

- **Cuisine Diversity:** The dataset shows a wide variety of cuisines, with North Indian and Chinese being the most frequent, while specialty cuisines often command higher average ratings.
- **Rating Distribution:** Most restaurants fall within the 3.0 to 4.0 range, suggesting that "Excellent" ratings (4.5+) are rare and highly localized.

- 

5. Strategic Recommendations (Alfido Tech Style)

Based on the data, here are 5 strategic recommendations for a platform like Alfido Tech:

1. **Hyper-Local Partnership:** Target the identified "Hotspot" cities for exclusive membership launches, as these areas have the highest user engagement potential.
2. **The "Value King" Badge:** Implement an algorithm to highlight restaurants with high ratings (4.0+) but low cost-for-two, appealing to the student and young professional demographic.
3. **Cuisine-Based Events:** Host "Food & Tech" networking mixers in cafes that serve the most popular cuisines identified (e.g., North Indian/Cafe culture).
4. **Trend Forecasting:** Use the "Cuisine vs. Rating" data to advise new restaurant partners on which niche cuisines are currently underserved but highly rated.
5. **Digital Integration:** Alfido Tech should integrate a "Mood-Based" search (e.g., 'Work-friendly', 'Quick Bite') based on the review keywords identified in the WordCloud analysis.